Acoustic Localization of Pipeline Robots

Submitted By

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Under the guidance of

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CERTIFICATE

This is to certify that the thesis entitled **"Acoustic Localization of Pipeline Robots"** submitted by **Gunda Sai Venkat** to the Indian Institute of Technology, Madras for the award of the degree of Dual Degree in Mechanical Engineering(Product Design) is a bonafide record of project work carried out by him under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Abstract

Key Words: *pipeline robots, localization, acoustic guided waves*

Pipeline inspection robots have increasingly become an important tool to ensure the safe, robust, and uninterrupted operation of pipelines in major oil and gas industries as well as in water supply lines. Accurately determining the location of these robots inside the pipelines is, therefore, a crucial process. Conventional methods like Global Positioning System (GPS) for real-time localization do not work in these cases due to the pipes' remoteness, either underground or underwater. This project focuses on overcoming these difficulties through the use of guided acoustic waves for localization. These waves have the advantage of minimum attenuation over large distances. The proposed method, along with Inertial Measurement Units (IMU) based dead reckoning data, can provide accurate real-time localization of pipeline robots.

Contents

Certificate **i**

Acknowledgements **iii**

Abstract **iv**

List of Figures **vii**

List of Tables **viii**

List of Symbols **ix**

1 Introduction **1**

1.1 Introduction 1
1.2 Problem Definition 2
1.3 Objectives 3
1.4 Methodology 3
1.5 Scope of the Thesis 4

2 Literature Review **6**

2.1 Localization Techniques 6
2.2 Acoustic Guided Waves 6

3 Calculation of Dispersion Curves **8**

3.1 Dispersion Curves 8
3.2 Formulation 8
3.3 Solving the Characteristic Equation 10
3.4 Results 15
3.5 Issues 15

4 FEA Simulations **16**

4.1 Simulation Setup 16
4.2 Validation of Velocities 17
4.3 Impinging Mode Shapes 18
4.4 Exciting Higher Velocity Modes 19

4.5 Attenuation Measurements 20
4.6 Simulations with Varying Operating Conditions 22
4.6.1 Applying Load on the Flat Surface of the Pipe 22
4.6.2 Applying Load on the Curved Surface of the Pipe 22
4.6.3 Applying Load on a Point Surface on the Pipe 23
4.6.4 Sensor Measurements on the surface of the Pipe 23
4.6.5 Selective Modal Excitation 24
4.7 Summary 25

5 Experimental Testing **26**
5.1 Experimental Setup 26
5.1.1 Pipe Assembly 26
5.1.2 Electronics 26
5.2 Testing 27
5.3 Discussion 29

6 Future Work **30**
6.1 Testing 30
6.2 Additional Simulations 30
6.3 Final Product Design 30

7 References **31**

List of Figures

1 Pipelines and Pigs 1
2 Cylindrical Coordinate System used for dispersion curves 8
3 Verification of Dispersion Curves 11
4 Verification of dispersion curves for 16 inch pipe using DISPERSE . . . 12
5 Phase Velocity vs ω plots 13
6 Group Velocity vs ω plots 14
7 Excess solutions in the dispersion curves 15
8 Assembly in ABAQUS 16
9 Received signal at d = 6m 17
10 Results for Frequency = 1000Hz 18
11 Results for Frequency = 2500Hz 19
12 Excitation of higher velocity modes 20
13 Amplitude vs Distance plots at various frequencies 21
14 Load on Flat Surface 22
15 Load on Curved Surface 23
16 Load on Point Surface 23
17 Signal when sensor is on the pipe 24
18 Selective Modal Excitation 25
19 Pipe Setup 26
20 Electronic Architecture 27
21 Received Signal Outside 27
22 Received Signal Inside 28
23 Microphones Used 29

List of Tables

1 Various Localization methods and associated problems 2
2 Material Properties 12
3 Calculated Attenuation Values 21

List of Symbols

$b =$ inner radii of the pipe

$d =$ inner radii of the pipe

$\omega =$ frequency

$\phi =$ scalar displacement potential

$\psi =$ vector displacement potential

$C_C =$ compressive wave velocity

$C_s =$ shear wave velocity

$C_{0m} =$ Wave velocity of mth mode

$C_1 =$ Wave velocity in the fluid

$S_z^L =$ axial displacement

$S_r^L =$ radial displacement

$J_0 =$ Bessel function of 1st kind

$Y_0 =$ Bessel function of 2nd kind

$A, B, C, D =$ constants

Chapter 1

Introduction

1.1 Introduction

Modern industries use an extensive array of pipelines connecting various points along the supply chain from factories to homes. Pipelines have been extensively used in oil refineries, fluid transport industry especially water and gas, power plants, chemical processing etc. Global gas pipeline industry alone is worth around $2480.2 billion in 2019 [1]. Pipes span over hundreds of kms traversing both underwater and underground terrains. There is a need for constant maintenance of these pipes to ensure efficiency in the operations of the industry. Over time, pipes start to develop cracks, scaling issues, leaks and blockages. Assessing the integrity of these pipes regularly is therefore important. Owing to the diverse set of operating conditions, location and span of these pipes, inspection can become complex and expensive.

Early leak detection has now become an important area of concern for industries. Conventional methods for leak detection range from manual inspection to advanced acoustic technologies. Until recently, manual inspection was the widely used method in spite of its significant drawbacks. Manual inspection requires huge resources and is prone to human error, failure to detect smaller leaks and cracks and is also not pos-

(a) A typical gas pipeline connecting cities.

(b) Pigs used for pipeline inspection

Figure 1: Pipelines and Pigs

sible for difficult terrain. Another option has been the use of pigs but this is possible only for piggable pipelines. Acoustic methods using guided waves can also be used which can detect even small leaks from a single point but is limited by the range of detection. A recent approach has been the use of in-pipe robots coupled with acoustic leak detection. The free-swimming robots traverse through the entire pipe length, detect leaks while also being able to avoid obstacles. They offer the convenience of automatic and accurate detection with minimal human intervention. The robots can also be used to perform other functions on the pipe as required.

A major problem impeding these robots is the inability to localize them inside the pipe accurately. Pipelines are generally underground or underwater and mostly inaccessible. These environments limit the use of many conventional techniques. Table 1 details the list of common localization techniques and the associated problems when used for pipeline robots.

S. No	**Method**	**Major Problems**
1.	Tethering	The range of inspection will be heavily constrained. It is difficult to deal with bends.
2.	GPS	Pipelines are generally underground or underwater where GPS is not accurate.
3.	Visual Odometry	Computer Vision based techniques fail as pipes don't have distinguishable features for detection.
4.	Wheel Odometry	Most pipeline robots are propelled by the fluid pressure. Hence there will be large errors due to robot slip. Moreover, due to the integral nature of the errors, they pile up as distance increases.

Table 1: Various Localization methods and associated problems

1.2 Problem Definition

One method of interest could be the use of acoustic waves guided through the pipe structure for localizing the robot Sound waves at certain frequencies based on the pipe dimensions and contents can travel large distances with minimal attenuation

and noise. This property can be exploited to localize robots inside pipes. The distance of the robot can be calculated from the Time of Flight(TOF) measurements and velocity calculations. This method has the advantage of being efficient, cheap and can be set up easily on any existing pipe networks. These measurements coupled with IMU data from the robots can provide for robust localization of the robots.

This project focuses on developing an acoustic localization method for accurate determination of the robot's location using Time of Flight (TOF) measurements. Sound waves at a particular frequency based on the pipe dimensions and contents can travel large distances at high velocities with minimal dispersion and noise. This, coupled with dead reckoning data from IMU's can provide robust localization of in-pipe robots.

1.3 Objectives

The main objective of the project is to develop an acoustic sensor system complete with a transmitter and receiver to accurately determine the location of pipeline robots. To maximise the range of working with minimal dispersion and attenuation, the ideal frequency and mode of propagation has to be calculated. The entire project has been structured into multiple objectives as follows:

1. Calculate the dispersion curves for the pipe-fluid system
2. Based on the group velocities, mode of propagation and attenuation characteristics, identify the ideal frequency for localization
3. Perform FEA simulations to validate the results from dispersion curves and identify the range, noise levels and other features of the acoustic propagation.
4. Perform experimental verification of the simulation results after developing the complete sensor architecture.
5. Design and develop a complete setup that can be reliably used in industries.

1.4 Methodology

Wave propagation in pipes is heavily dependent on the dimensions and material of the pipe as well as the fluid inside. Dispersion curves can be calculated by solving the

wave equations coupled with boundary conditions based on the pipe fluid interaction. From the dispersion curves, a suitable frequency and mode can be selected that has the following desirable characteristics

1. Has a high group velocity that is different from other modes.
2. Has very less dispersion
3. Should be easily excitable
4. Should have least attenuation

Using the selected frequency and mode, FEA simulations are performed using ABAQUS to calculate range and validate the velocities. Acoustic pressure is applied to the fluid at one end and the system response at multiple points measured. Based on the amplitude and time of arrival of the acoustic waves, the range and velocities are calculated and validated with the dispersion curves.

Testing will be conducted on a pipe setup using the required electronics and software. The preliminary testing will be on a smaller assembly which can then be expanded to larger setups. Dead reckoning data from IMU's will also be incorporated for accurate localization. Based on the testing results and after suitable improvements, the solution will be developed into a final product to be used in industries.

1.5 Scope of the Thesis

This thesis is an attempt to summarize the understandings developed from various simulations and experimental investigations. A detailed analysis and results have been presented over the course of the report to study the feasibility of acoustic localization. The thesis organization is as follows:

Chapter 1 deals with the introduction and basic overview of the project. The objectives and methodology to be employed are also discussed.

Chapter 2 presents a literature review of the existing technologies related to localization of in-pipe robots. Latest research on acoustic guided waves and their properties

are also presented.

Chapter 3 details the calculation of dispersion curves to be used for the generation of the acoustic waves. The desired frequency and mode are selected through these dispersion curves.

Chapter 4 provides detailed results on the FEA simulations performed on ABAQUS. Several different simulations with varying operating conditions have been conducted and their results presented.

Chapter 5 presents the results from the experimental testing that has been performed on a pipe setup.

Chapter 6 provides the scope for future work and research for the above problem statement.

Chapter 2

Literature Review

2.1 Localization Techniques

Localization of in-pipe robots has become and important emerging problem in the field of pipeline inspection and maintenance. Accordingly, there is research available that tries to develop new technologies that can aid in the process. Several methods ranging from SLAM(Simultaneous Localization and Mapping) techniques to ultrasonic pingers have been employed with limited success.

Yoshiaki Bando et al.,[2] uses a speaker and microphone system to localize an in-pipe robot with an error of 7%. However, it suffers from low range and inability to work in fluid filled pipes. Richard Fletcher et al.,[3] showcase an ultrasonic wave based technique coupled with IMU information used in Smart Ball developed by Pure Technologies. Carlos Rizz et al.,[4] used Radio Frequency waves for robot localization but it suffered from high attenuation rates.

Rob Worley et al.,[5] used a system of acoustic signatures where the spatial data was captured based on pipe's response to acoustic waves at each location.Ke Ma et al.,[6] developed SLAM algorithms which uses pipe vibrations to develop a map for localization. Both these methods are difficult to achieve due to the featureless environment inside pipes. Jonathan Aitken et al.,[7] gives an overview of existing SLAM methods and the challenges in localizing in-pipe robots. Zheng Liu et al.,[8] describes the existing technologies and scope for future development in inspection as well as localization.

2.2 Acoustic Guided Waves

Lafleur et al.,[9] provides detailed theoretical analysis of axisymmetric wave propagation in fluid filled pipes. The paper develops the required formulation to model duct guided waves. The dissertation by Del Grosso[10] also contains the equations and their solutions for different types of wave propagation in pipes with varying operating and

boundary conditions. The thesis by Jian Ma[11] gives applications of guided ultrasonic waves for measurements inside pipes.

There has been considerable research in the field of acoustic guided waves. But none of it focuses on using it solve the localization problems faced by in-pipe robots. Acoustic guided waves satisfy many of the requirements for robot localization and therefore offers and enterprising solution for the same.

Chapter 3

Calculation of Dispersion Curves

3.1 Dispersion Curves

Dispersion curves are a set of curves that represent the propagation of wave modes that are found in a specific geometry. When acoustic waves propagate through fluids, there exist several modes of propagation depending on the frequency of the wave. Each mode corresponds to a different wave speed and attenuation characteristics. Dispersion curves help in identifying all such modes, frequencies and speeds of wave propagation possible. They are calculated by solving the wave equations coupled with boundary conditions at the pipe-fluid interface.

3.2 Formulation

Lafleur et al.,[9] and Del Grosso[10] provide a detailed theoretical framework and analysis for describing axisymmetric wave propagation in fluid filled pipes. The characteristic equation for solving the model dispersion relations is derived theoretically. A cylindrical elastic tube of inner and outer radii b and d with fluid inside is considered. Axisymmetric wave of frequency ω is assumed to travel along the pipe.

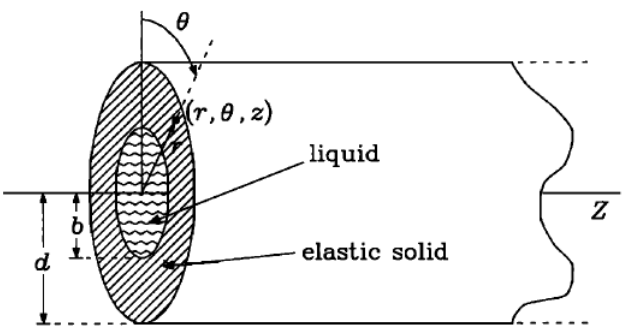


Figure 2: Cylindrical Coordinate System used for dispersion curves

The longitudinal and shear wave equations for the fluid are given by

$$\nabla^2\phi - \frac{1}{C_c^2}\frac{\partial^2\phi}{\partial t^2} = 0 \tag{1}$$

$$\nabla^2\psi - \frac{\psi}{t^2} - \frac{1}{C_s^2}\frac{\partial^2\phi}{\partial t^2} = 0 \tag{2}$$

The axial and radial components of particle displacement in the fluid are given by

$$S_z^L(r,z,t) = i\phi_0 q_{0m} J_0(rX_{0m}/b)e^{i(q_{0m}z-\omega t)} \tag{3}$$

$$S_r^L(r,z,t) = -(\phi_0 X_{0m}/b)J_1(rX_{0m}/b)e^{i(q_{0m}z-\omega t)} \tag{4}$$

Similarly, the components in pipe wall are

$$S_z^W(r,z,t) = iq_{0m}[AJ_0(rP_m) + BY_0(rP_m)] + i(q_{0m}[CJ_0(rT_m) + DY_0(rT_m)]e^{i(q_{0m}z-\omega t)} \tag{5}$$

$$S_r^W(r,z,t) = -P_m[AJ_1(rP_m) + BY_1(rP_m)] - i(q_{0m}[CJ_1(rT_m) + DY_1(rT_m)]e^{i(q_{0m}z-\omega t)} \tag{6}$$

Here,

$$X_{0m} = b\sqrt{k_1^2 - q_{0m}^2} \qquad P_m = \sqrt{k_c^2 - q_{0m}^2} \qquad T_m = \sqrt{k_s^2 - q_{0m}^2} \tag{7}$$

$$q_{0m} = \omega/C_{0m} \qquad k_c = \omega/C_c \qquad k_s = \omega/C_s \qquad k_1 = \omega/C_1 \tag{8}$$

The boundary conditions are as follows

1. The liquid is assumed to be without shear. Hence tangential stress at the elastic boundary of the pipe should vanish.

$$\sigma_{rz}|_{r=b} = 0 \tag{9}$$

2. Normal stress in the elastic solid is equal to the negative fluid pressure at the boundary

$$P|_{r=b} = -\sigma_{rr}|_{r=b} \tag{10}$$

3. There is a continuity of displacement at the interface in both fluid and solid.

$$S_r^L|_{r=b} = S_r^W|_{r=b} \tag{11}$$

Combining all the above equations and solving, we get the final characteristic equation for the dispersion curves relating ω and q_{0m}.

$$\begin{aligned}
1 + [L_{11}(P_m)L_{00}(T_m)](\frac{\pi^2 q_{0m}^2 bdP_m^2 T_m^2}{8E_m^2}) + [L_{11}(T_m)L_{00}(P_m)](\frac{\pi^2 bdE_m^2}{8q_{0m}^2}) + \\
[L_{10}(P_m)L_{01}(T_m) + L_{01}(P_m)L_{10}(T_m)](\frac{\pi^2 bdP_m T_m}{8}) + \\
[bL_{11}(P_m)L_{10}(T_m) + d(1+Q_m b)L_{11}(P_m)L_{01}(T_m)](\frac{\pi^2 P_m^2 T_m}{8E_m} - \frac{\pi^2 q_{0m}^2 P_m^2 T_m}{8E_m^2}) + \\
[bL_{11}(T_m)L_{10}(P_m) + d(1+Q_m b)L_{11}(T_m)L_{01}(P_m)](\frac{\pi^2 P_m E_m}{8q_{0m}^2} - \frac{\pi^2 P_m}{8}) + \\
[(1+Q_m b)L_{11}(T_m)L_{11}(P_m)](\frac{\pi^2 P_m^2}{8q_{0m}^2} + \frac{\pi^2 P_m^2 q_{0m}^2}{8E_m^2} - \frac{\pi^2 P_m^2}{4E_m}) = 0
\end{aligned} \tag{12}$$

Here

$$E_m = q_{0m}^2 - k_s^2/2 \tag{13}$$

$$Q_m = \frac{\rho_L \omega^2 b J_0(X_{0m})}{2\rho_W C_s^2 X_{0m} J_1(X_{0m})} \tag{14}$$

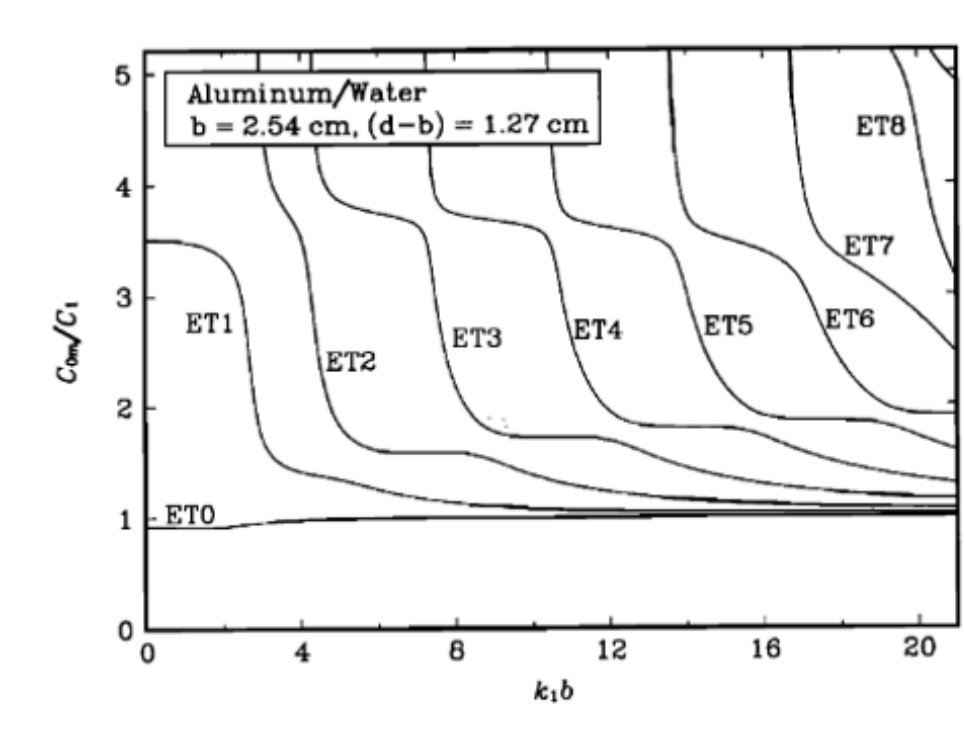
$$L_{mn}(y) = J_m(dy)Y_n(by) - J_n(by)Y_m(dy) \tag{15}$$

Note: All the symbols used in the above equations have been specified in List of Symbols

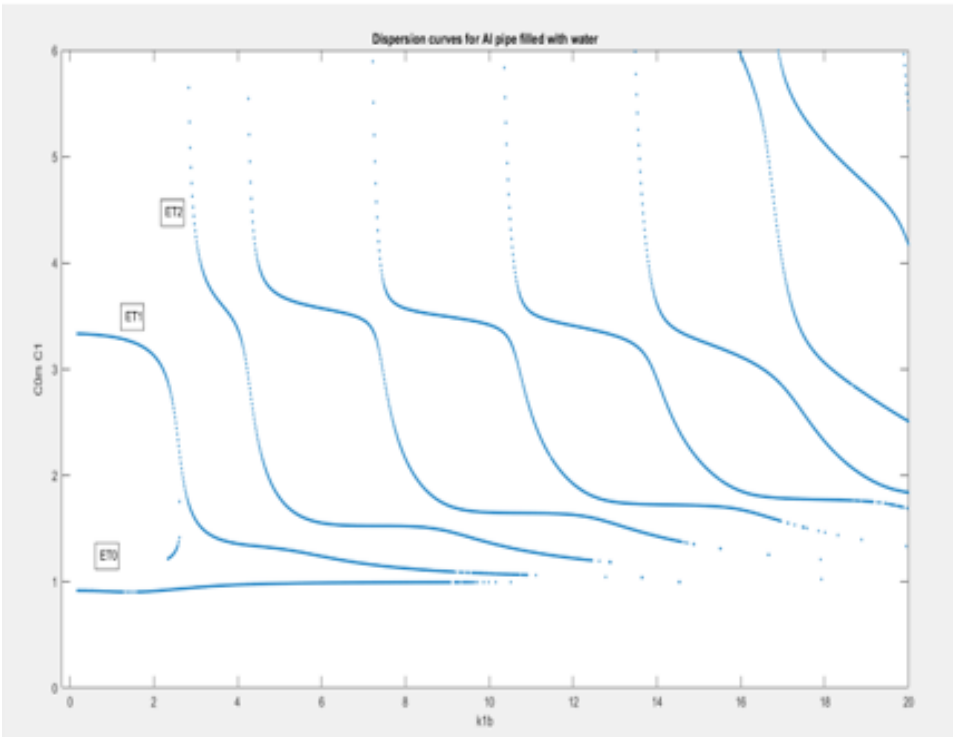
3.3 Solving the Characteristic Equation

The above characteristic equation was solved using MATLAB. Due to the complexity of the equation, fsolve from MATLAB was used to solve it numerically. The initial values of C_{0m} are taken in steps from 0 to a high value and for each initial C_{0m} and ω, the equation was solved individually. All the different solutions were plotted for the dispersion curves. To verify the correctness of the code, the dispersion curves were generated and compared with those from Lafleur et al.,[9] for the same pipe

dimensions and material. As observed in Figure 3, the plots match with each other verifying the developed code.



(a) Dispersion Curves from the Lafleur paper



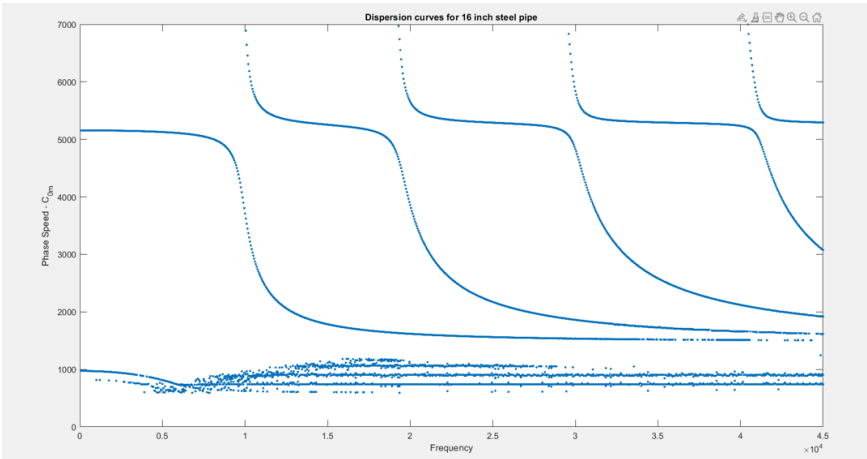
(b) Calculated Dispersion Curves

Figure 3: Verification of Dispersion Curves

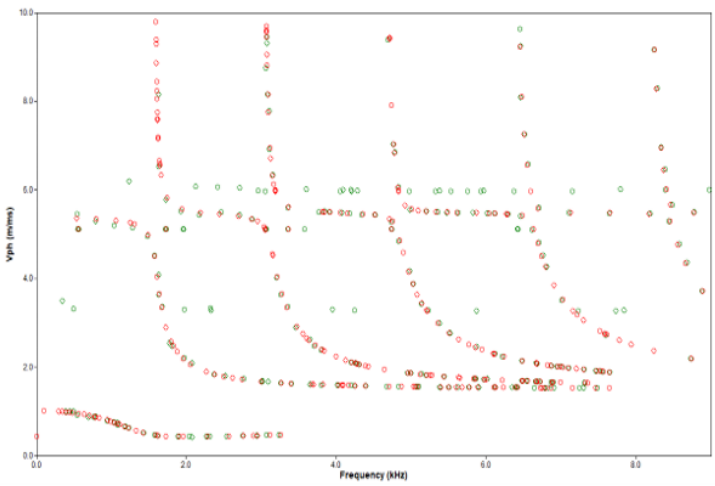
For the preliminary analysis, dispersion curves were generated for steel pipes of 10, 16 and 24 inch diameters with 7mm thickness. The material properties used for the analysis are presented in Table 2. Both phase and group velocity curves vs frequency have been plotted. These plots were further verified with those generated using the software DISPERSE.

Material Properties			
Pipe		**Water**	
Material	Steel	**Material**	Water
Density	7850 kg/m^3	**Density**	1000 kg/m^3
$\mathbf{C}_l$	5921 m/s	$\mathbf{C}_1$	1481 m/s
$\mathbf{C}_s$	3091 m/s		

Table 2: Material Properties

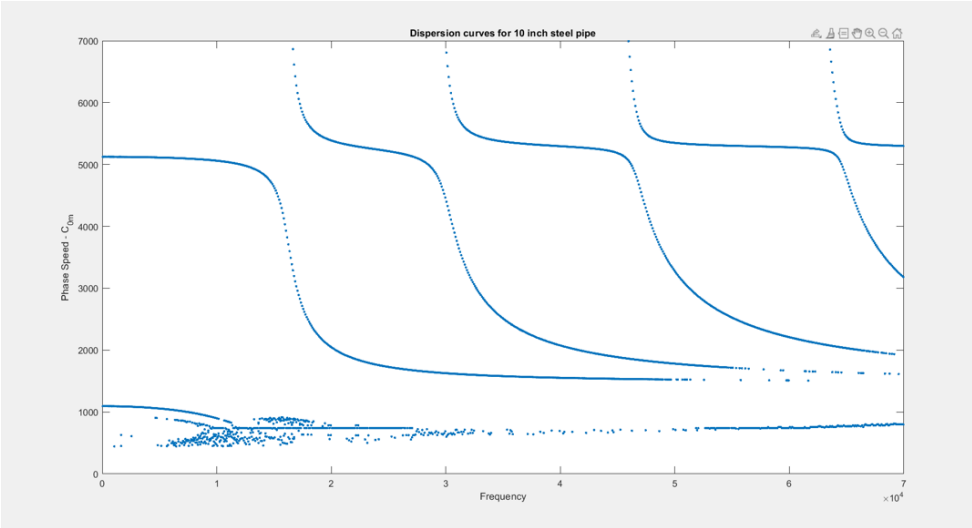


(a) Phase Velocity vs ω - Calculated

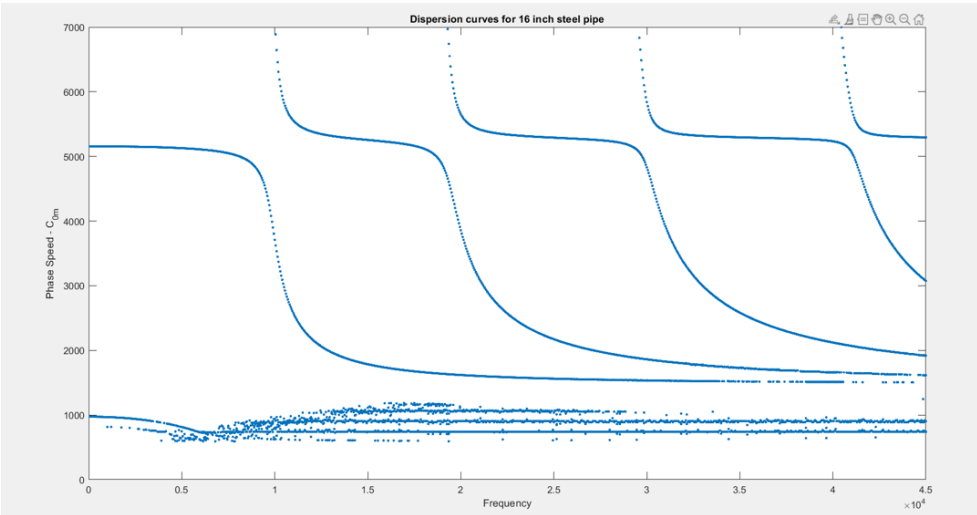


(b) Phase Velocity vs ω - DISPERSE

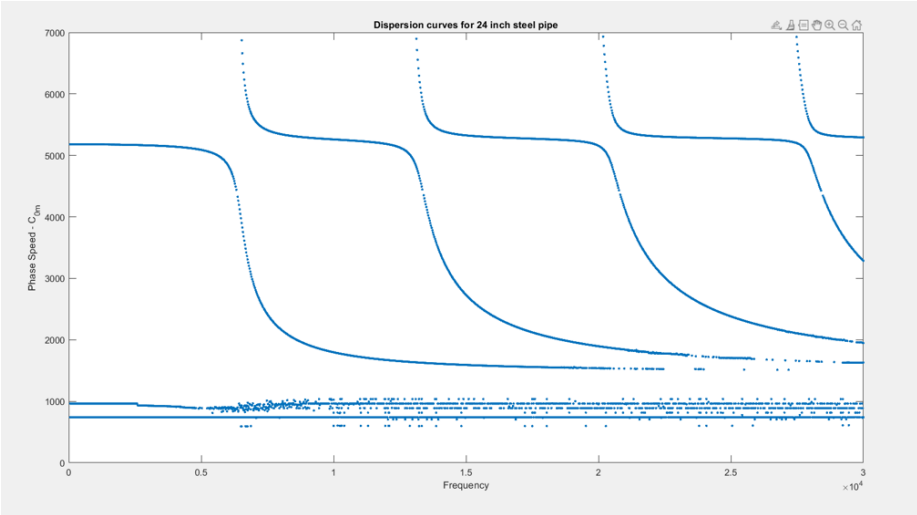
Figure 4: Verification of dispersion curves for 16 inch pipe using DISPERSE



(a) 10 inch pipe

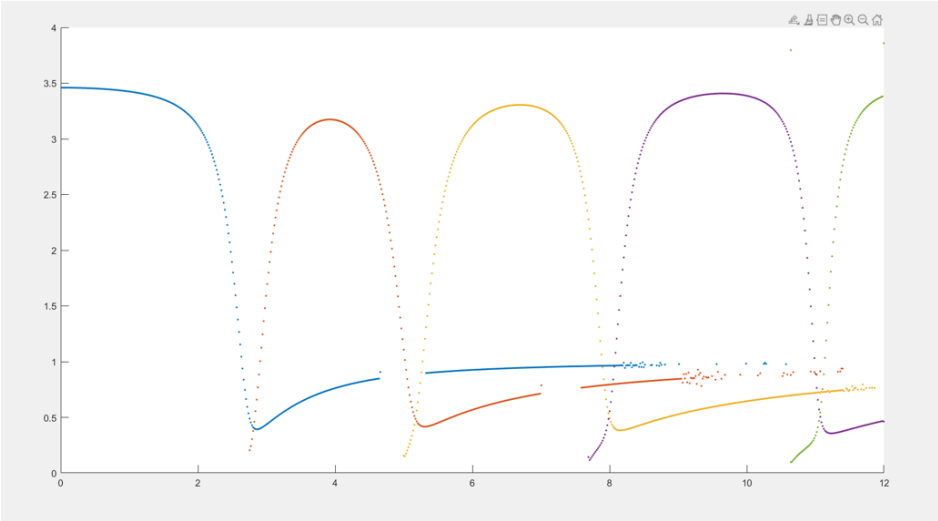


(b) 16 inch pipe

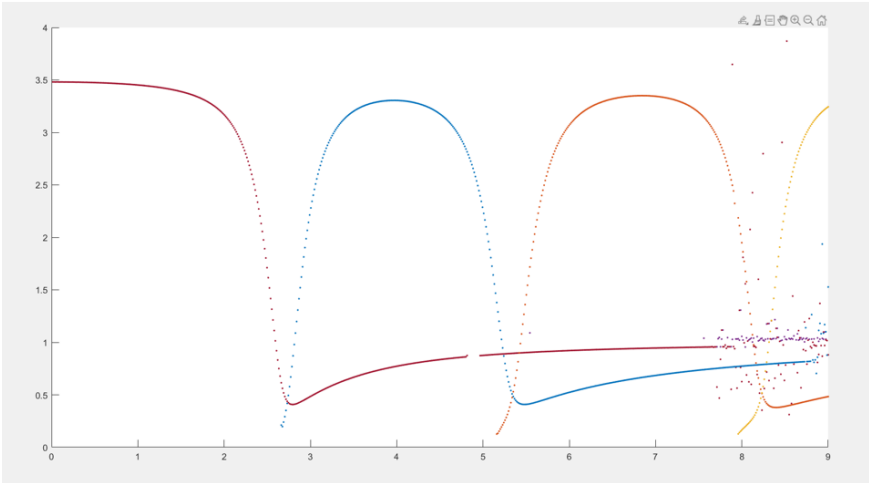


(c) 24 inch pipe

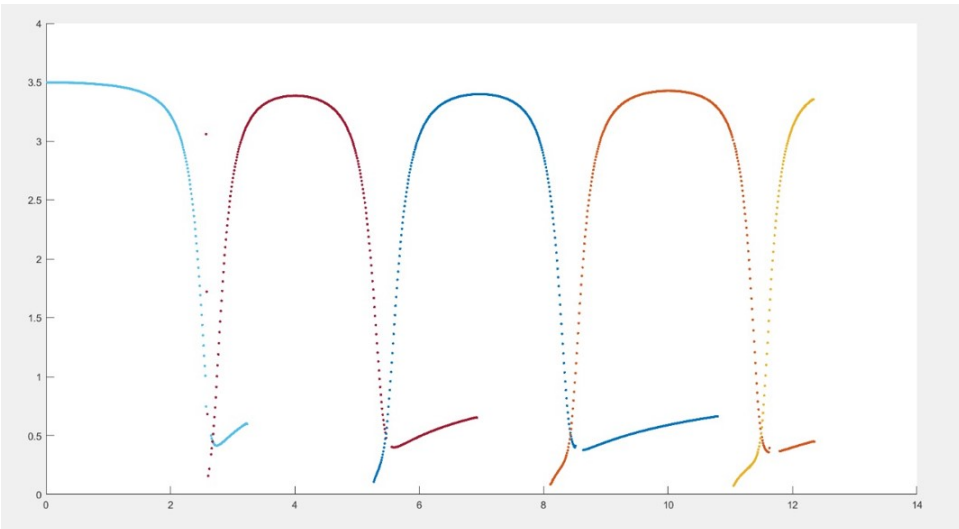
Figure 5: Phase Velocity vs ω plots



(a) 10 inch pipe



(b) 16 inch pipe



(c) 24 inch pipe

Figure 6: Group Velocity vs ω plots

3.4 Results

The dispersion curves have been validated by the Lafleur paper and DISPERSE software. For the steel pipes, it can be seen that as frequency increases, the number of modes at each frequency increase. The group velocity plots show that for every mode, there exist a band of frequencies at which the velocity is highest and is stable. Exciting the system at this frequency will lead to faster wave propagation and with less dispersion. The velocity is also significantly different from other modes which will help in isolating the signal easily. All the modes have similar high group velocities but the ET0 mode is preferred as there are no other interfering modes at this frequency range. These results will be further verified through FEA(Finite Element Analysis) simulations to gather further data to decide on a suitable exciting frequency and mode.

3.5 Issues

In the calculated dispersion curve plots, when $C_{0m} < C_1$, there occur multiple solutions which are not valid, This happens when thickness of the pipe is very less when compared with the diameter. Figure 7 shows the excess points in the plot that are invalid. This issue occurs due to the tolerance of the solver used in MATLAB(fsolve). When $C_{0m} < C_1$, the characteristic function value is close to 0 for all values of C_{0m}. Hence, for each initial value of C_{0m}, the solution is found at that point itself causing errors.

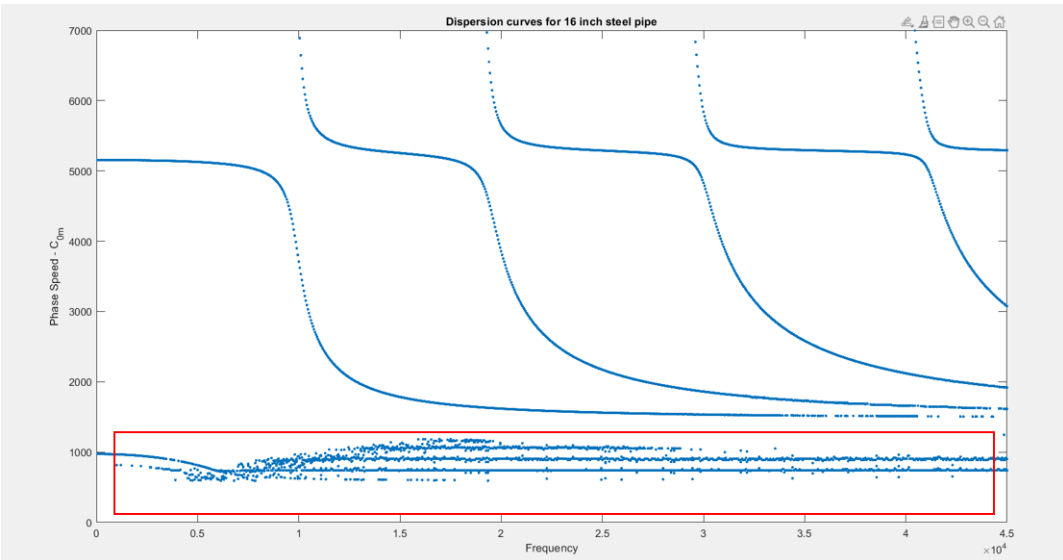


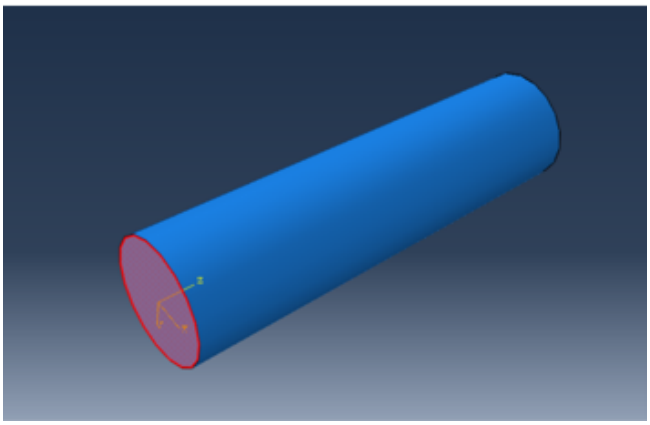
Figure 7: Excess solutions in the dispersion curves

Chapter 4

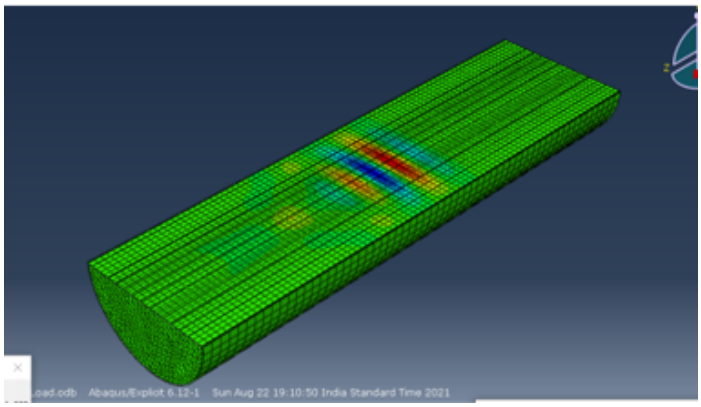
FEA Simulations

4.1 Simulation Setup

Finite Element Analysis(FEA) simulations for this system have been performed on ABAQUS to further validate the dispersion curves as well as gain additional information with respect to attenuation, signal to noise ratio etc. 16 inch pipe with 7mm thickness has been modelled and the material properties have been assigned. The length of the pipe was chosen according to convenience. The pipe and fluid were designed as two independent parts and later assembled. The fluid was meshed using AC3D4 elements while the pipe was meshed using C3D8R elements. The mesh size was chosen such that it lies between $\lambda/8$ and $\lambda/20$. Dynamic, Explicit analysis was performed using automatic time incrementation. The load was applied as acoustic pressure boundary condition on one end of the fluid surface with the amplitude varying according to the frequency of vibration. Non reflecting interactions were applied at the other end of the pipe and fluid. Tie constraint was applied at the fluid pipe interface.



(a) System Assembly in ABAQUS

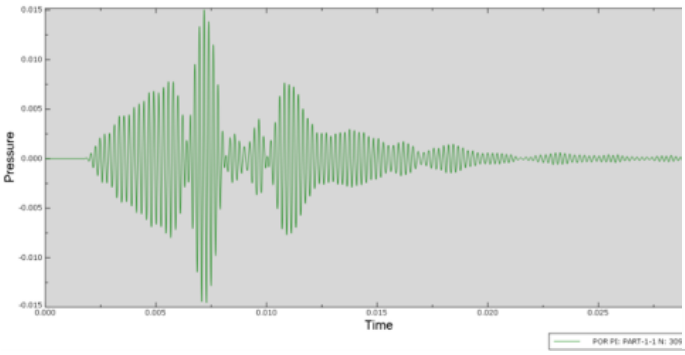


(b) Section View of the meshed assembly

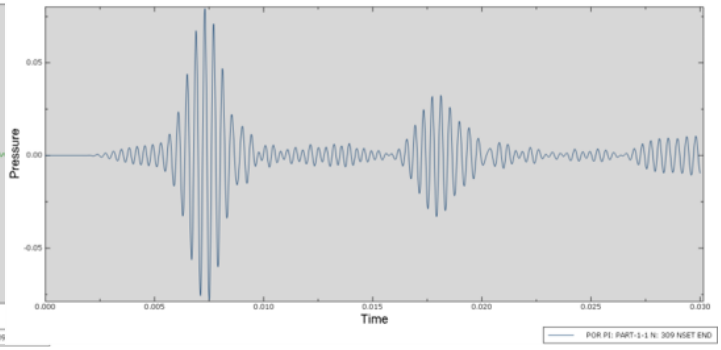
Figure 8: Assembly in ABAQUS

4.2 Validation of Velocities

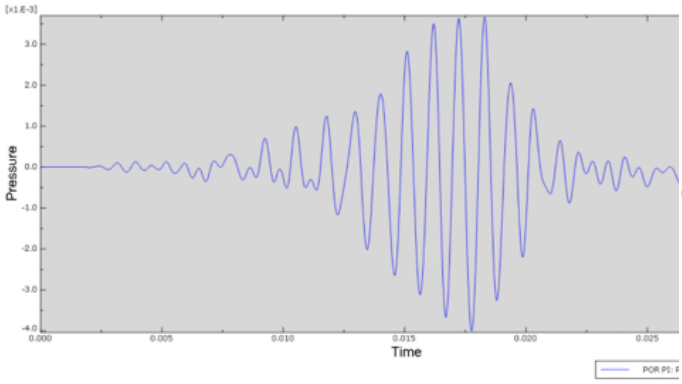
The group velocities calculated using dispersion curves were initially validated through simulations. A 6m long pipe was considered for the analysis. Acoustic pressure was applied at different frequencies and the wave propagation was observed at multiple points along the pipe separated by 1m. The velocity was calculated by the time taken by the wave to reach the end of the pipe and compared with the values from dispersion curves.



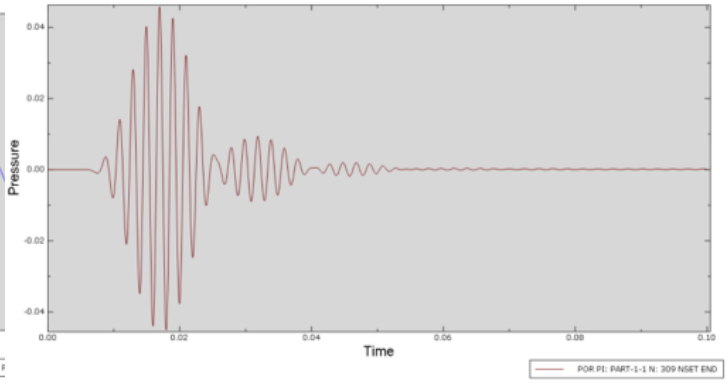
(a) Frequency = 5000Hz



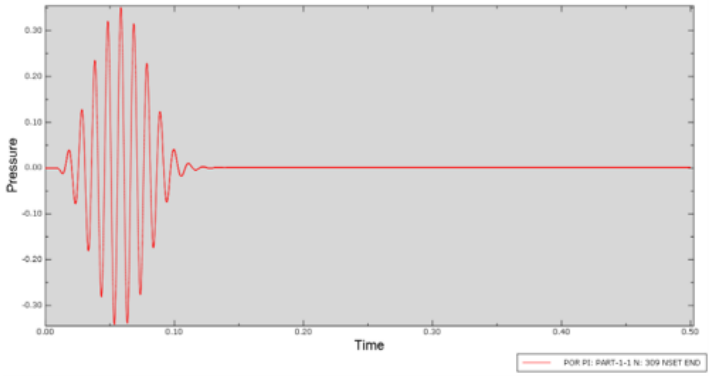
(b) Frequency = 2500Hz



(c) Frequency = 1000Hz



(d) Frequency = 500Hz



(e) Frequency = 100Hz

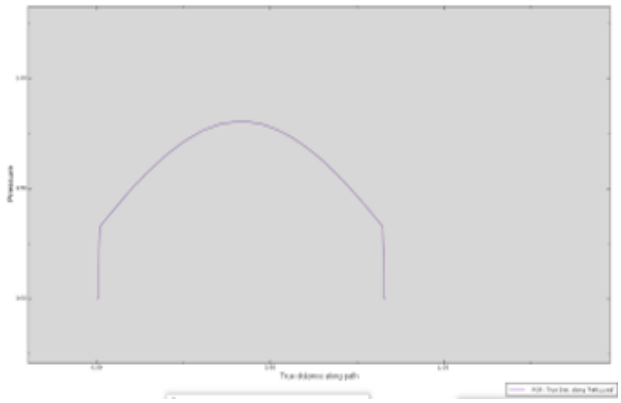
Figure 9: Received signal at d = 6m

Figure 9 shows the signal received at the end of the pipe for different frequencies.

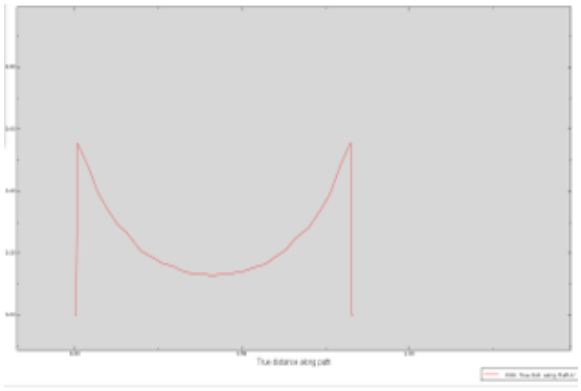
It can be seen that with higher frequencies, multiple modes exist with varying velocities. The velocity for each mode has been compared with the calculated values from dispersion curves. Both the values seem to match closely thus validating the simulations. It has been observed that in most cases, only one mode dominates the received signal while the others are very faint or absent. For all the frequencies, it is seen that the mode with highest group velocity never gets excited and it is only the mode with lower velocities that are seen.

4.3 Impinging Mode Shapes

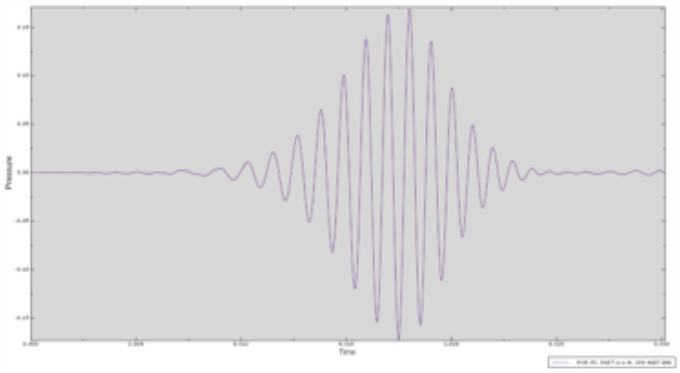
In the above simulations, the acoustic pressure was applied throughout the surface of the fluid at one end uniformly. It was observed that the higher group velocity mode didn't get excited. To specifically excite this particular mode for its high velocity, the acoustic pressure load was applied on the surface according to the mode shape of that particular mode. The mode shape was found using DISPERSE. This shape was applied at one end of the fluid through a special function in ABAQUS.



(a) Applied Mode Shape

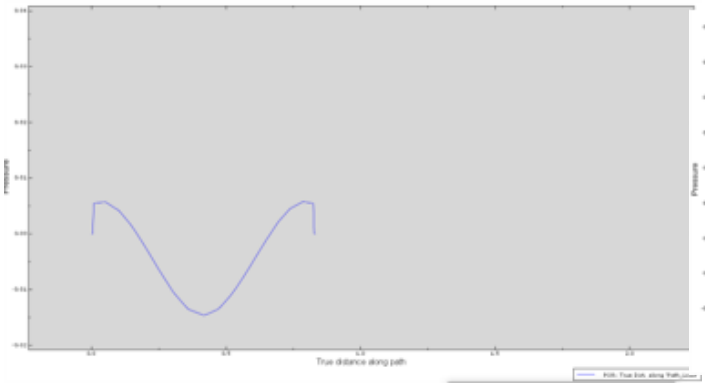


(b) Received Mode Shape

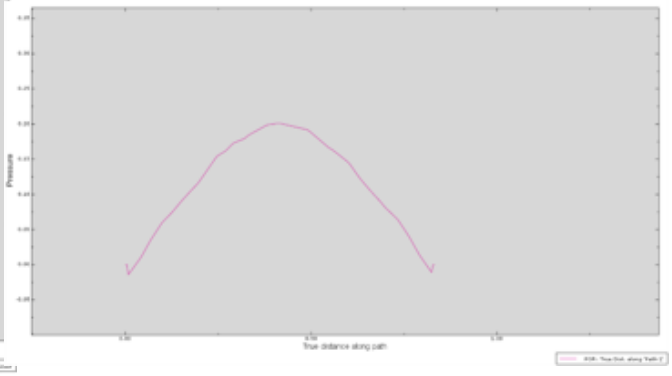


(c) Received signal at d = 6m

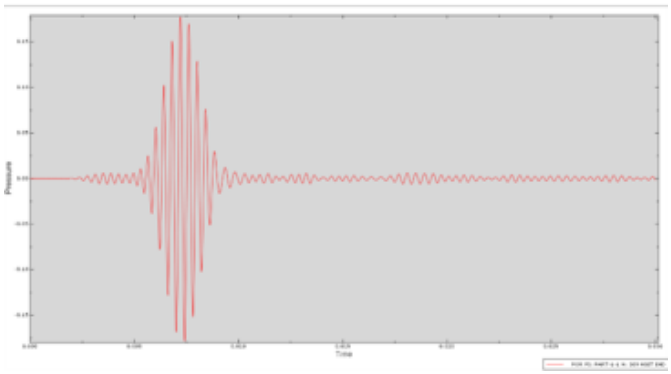
Figure 10: Results for Frequency = 1000Hz



(a) Applied Mode Shape



(b) Received Mode Shape



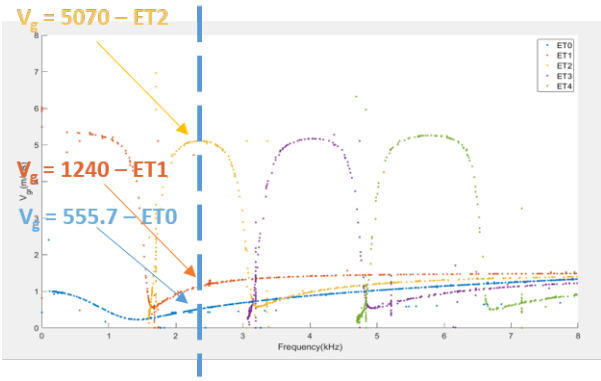
(c) Received signal at d = 6m

Figure 11: Results for Frequency = 2500Hz

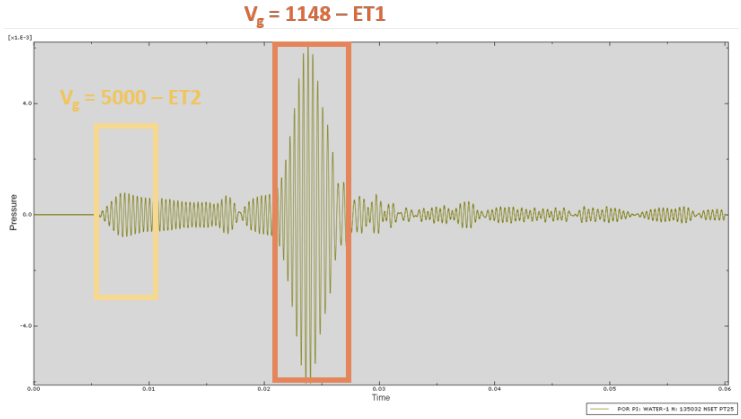
Figures 9 and 10 show the results for the frequencies 1000Hz and 2500Hz. It has been observed that the results did not change even after impinging the mode shapes. The lower velocity mode only got excited for each frequency.

4.4 Exciting Higher Velocity Modes

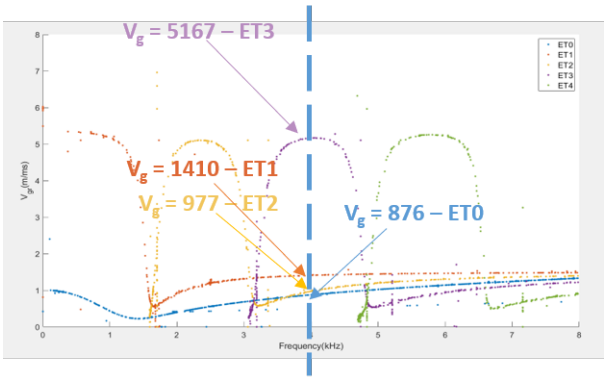
To analyze the reason for non-excitation for higher velocity modes, simulations have been performed on a longer pipe of length 25m. It has been observed that at far off distances, all the modes exist and can be distinguished from each other. At smaller distances, one mode dominates in amplitude and hence the other modes weren't easily discernible. Higher velocity modes also tend to lose a lot of their intensity immediately after application at the beginning. This leads to predominance of a single mode at the beginning but as distance increases, it has been observed that due to attenuation all the modes tend to be of similar amplitude and are received at the output.



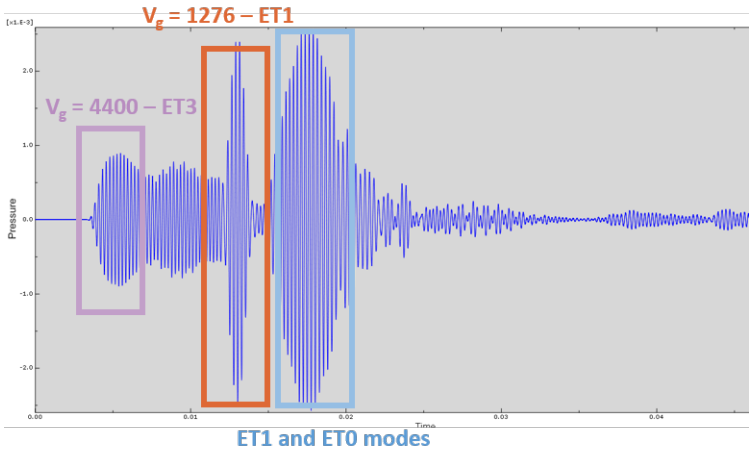
(a) Group Velocities and Modes at 2500Hz



(b) Received Signal at d = 25m for 2500Hz



(c) Group Velocities and Modes at 4000Hz



(d) Received Signal at d = 25m for 4000Hz

Figure 12: Excitation of higher velocity modes

4.5 Attenuation Measurements

To measure the attenuation of the signal, simulations have been performed on a 50m long pipe. The signal amplitude was measured at every 1m distance. Attenuation was calculated as

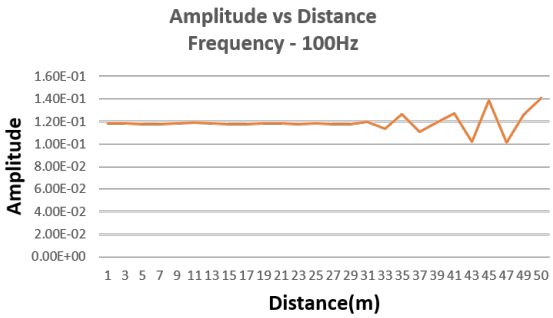
$$Attenuation = \frac{20\ log_{10}(\frac{P_B}{P_A})}{d_{AB}}$$

$$where,$$

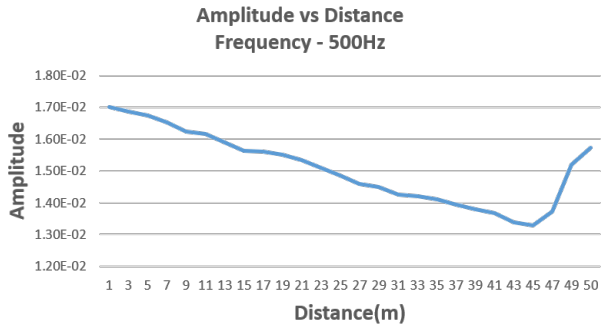
$$P_A = \text{Amplitude at point A}$$

$$P_B = \text{Amplitude at point B}$$

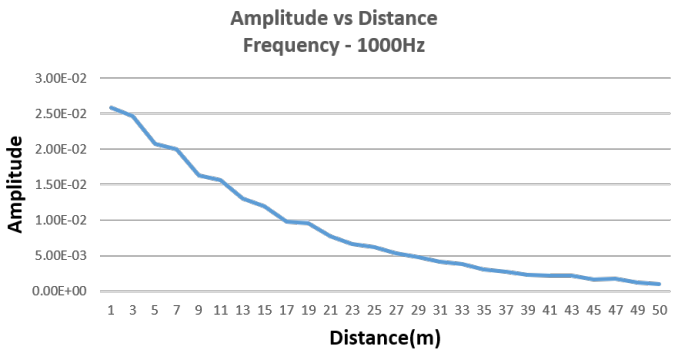
$$d_{AB} = \text{distance between the points A and B}$$



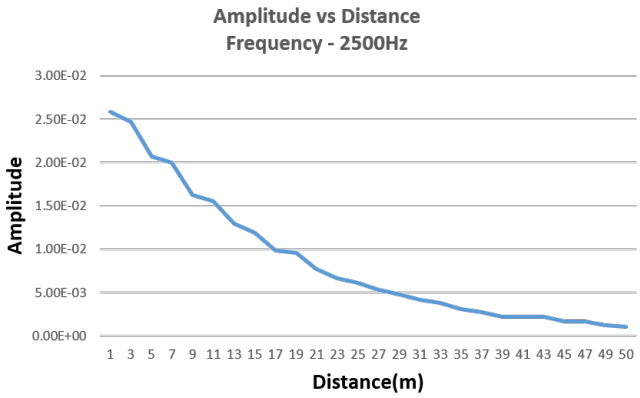
(a) Frequency = 100Hz



(b) Frequency = 500Hz



(c) Frequency = 1000Hz



(d) Frequency = 2500Hz

Figure 13: Amplitude vs Distance plots at various frequencies

Frequency(Hz)	**Attenuation(dB/m)**
100	-0.008
500	-0.0517
1000	-0.4580
5000	-0.5269

Table 3: Calculated Attenuation Values

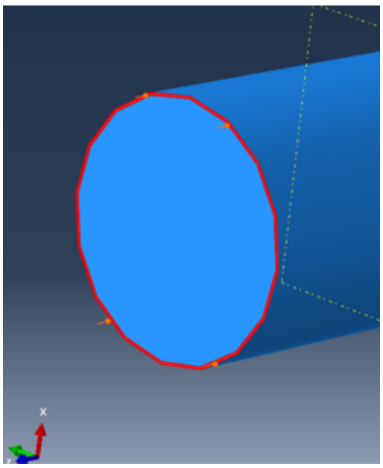
From the attenuation values it can be seen that attenuation increases with frequency. At 100Hz, the attenuation is very low and the wave can travel for considerable distance without any change in signal strength. In the amplitude vs distance plots in Figure 13, there exists a discrepancy at the end where the signal strength again tends to increase. This is because of the reflecting waves from the other end. Though non-reflecting boundary conditions were used, slight reflections still occurred leading to this anomaly.

4.6 Simulations with Varying Operating Conditions

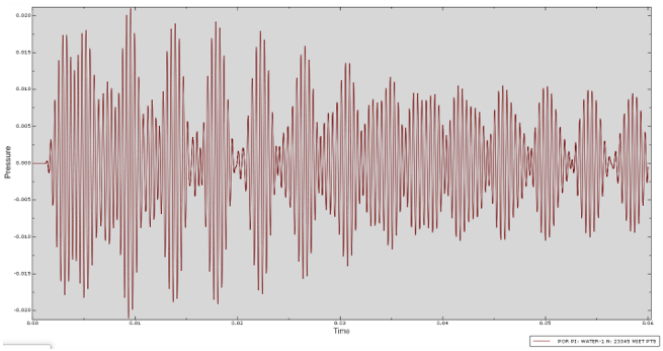
It was observed that applying the load and receiving the signal at the centre of the fluid might not be feasible in practical conditions. So simulations have been performed with varying operating conditions to assess their feasibility and practicability. All the following simulations were performed for the frequency of 2500Hz.

4.6.1 Applying Load on the Flat Surface of the Pipe

In earlier simulations, load was applied on one end of the fluid surface. In this case, load is applied on one end of the pipe surface instead of the fluid. All the other simulation parameters remain the same. The signal is again received at the center of the fluid surface on the other end.



(a) Position of Load Application



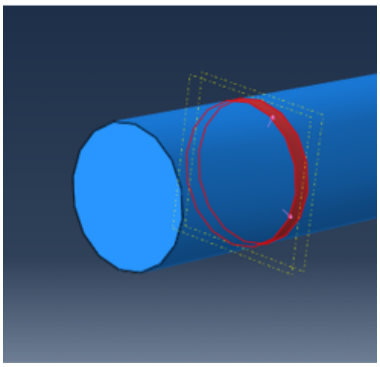
(b) Received signal

Figure 14: Load on Flat Surface

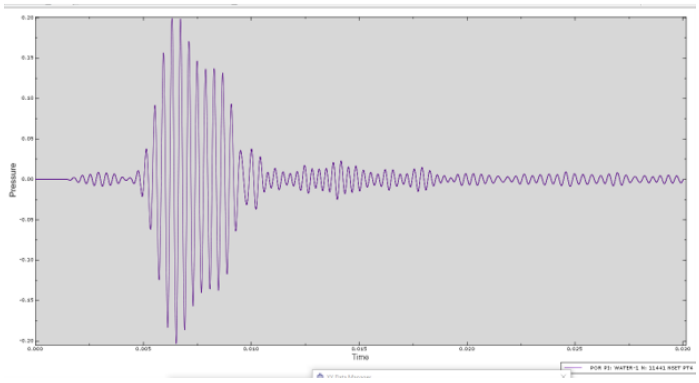
The velocity of wave propagation is very high in this case. But attenuation also increases. But applying load in this way might also be practically difficult.

4.6.2 Applying Load on the Curved Surface of the Pipe

In this case, the load is applied along the curved surface of the pipe. The results and group velocity are similar to those in the case where load is applied on the fluid surface.



(a) Position of Load Application

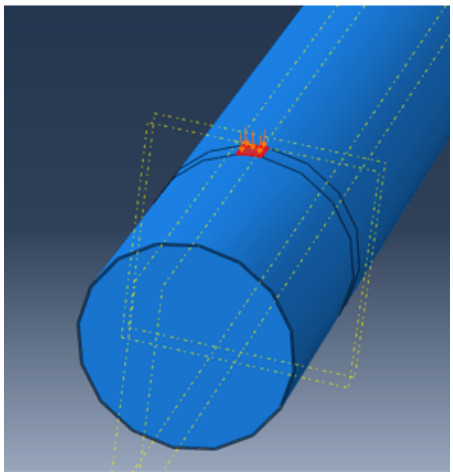


(b) Received signal

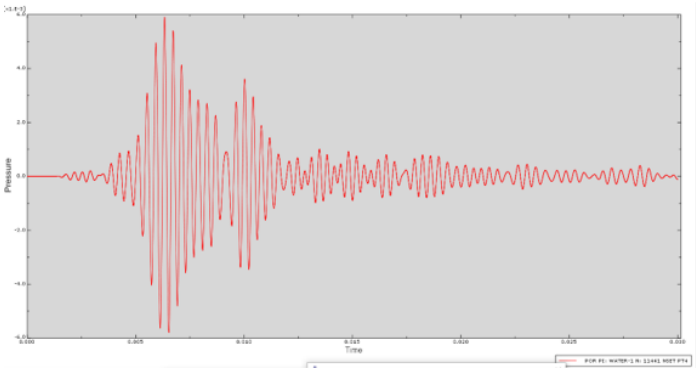
Figure 15: Load on Curved Surface

4.6.3 Applying Load on a Point Surface on the Pipe

In this case, the load is applied at a particular point on the curved surface of the pipe. The same mode gets excited even in this case but there also exists additional noise in the received signal.



(a) Position of Load Application

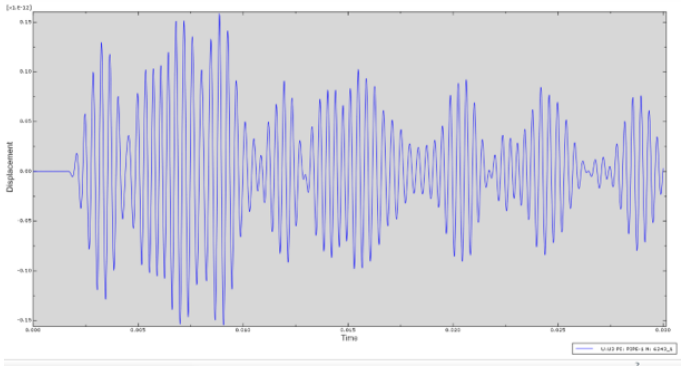


(b) Received signal

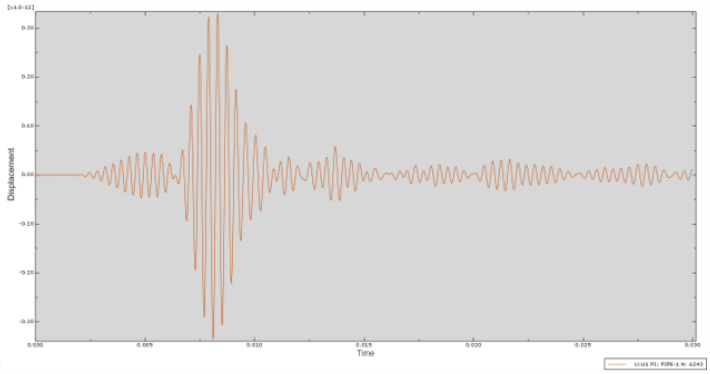
Figure 16: Load on Point Surface

4.6.4 Sensor Measurements on the surface of the Pipe

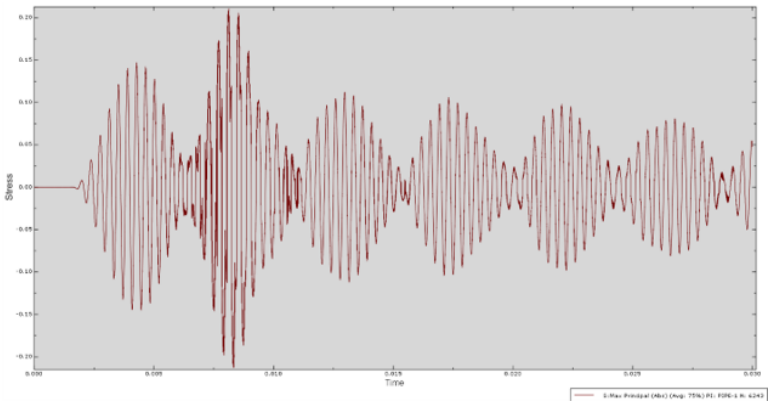
In this case, load is again applied at the centre of the fluid surface while the measurements are taken on the pipe surface. Stress and displacements are measured and plotted. It can be seen that the mode can be identified through this method as well.



(a) Axial Displacement



(b) Radial Displacement



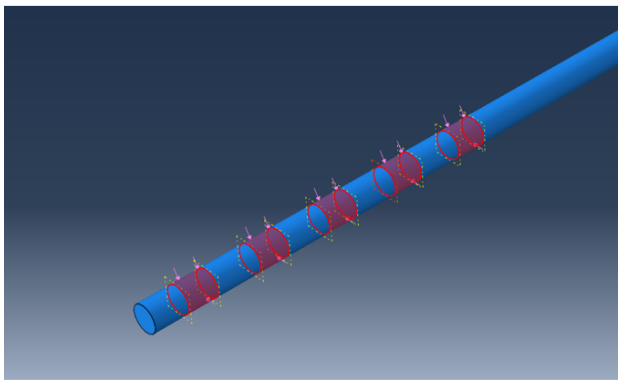
(c) Stress

Figure 17: Signal when sensor is on the pipe

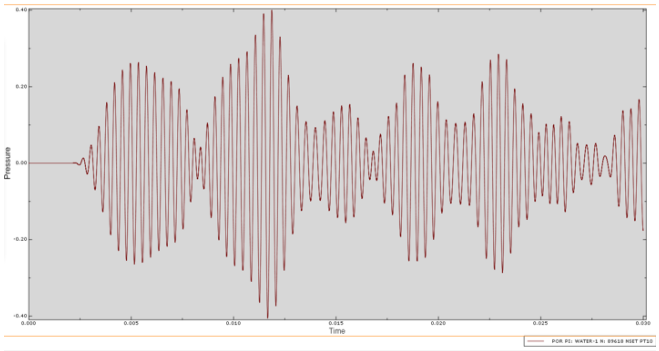
4.6.5 Selective Modal Excitation

As the earlier results show that there is a dominant mode with lower velocity that always appears in the signal, selective modal excitation was performed based on Dileep et al., [12] and Jacob et al., [13]. In this method, load is applied at multiple space points along the length of the pipe with the spacing to be equal to the wavelength of the required mode to be excited. This method would help in exciting only the required mode.

From the results from Figure 18, it is observed that this method does not work in this case. This could be because in this particular case, as the frequency is very low, the wavelength is very high. This means that the spacing between successive load application points is high.



(a) Position of Load Application



(b) Received signal

Figure 18: Selective Modal Excitation

4.7 Summary

Several FEA simulations have been performed to validate existing information and gain additional knowledge on the pipe fluid assembly, Dispersion curves and group velocities have been validated. Attenuation characteristics have also been identified. Several other simulations have been performed to assess different operating conditions. It has been observed that the load and the sensor could be placed even on the pipe to receive the signal. Further studies have to be performed on these cases for better understanding, The higher velocity modes have also been identified in longer pipes. Based on the group velocity and attenuation, 2500 HZ frequency was considered to be ideal for further experimental testing.

Chapter 5

Experimental Testing

5.1 Experimental Setup

5.1.1 Pipe Assembly

Experiments were conducted on a steel pipe setup.The pipe diameter is 12 inches while thickness was 6mm. The pipe length is 8m. Both ends of the pipe are U shaped. The speaker is placed at one end while the microphone was placed at the other end of the pipe. The pipe was completely filled with water. Figure 19 shows the images of the pipe used.

Figure 19: Pipe Setup

5.1.2 Electronics

The electronic architecture consists of ESP32 micro controllers, Real Time Clock(RTC) modules, INMP441 microphone and a speaker. The speaker is attached at one end of the pipe while the mic is attached to the other end. The speaker emits acoustic waves while the receiver captures them and calculates the time of flight. From the calculated velocities using dispersion curves and simulations, the distance can be measured. The electronics setup is interfaced through Arduino platform and the necessary codes for the same were developed.

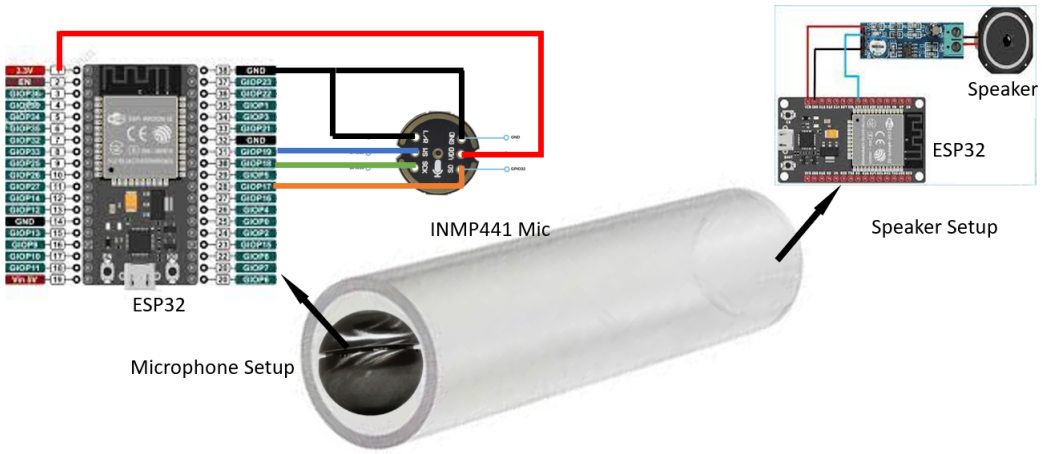
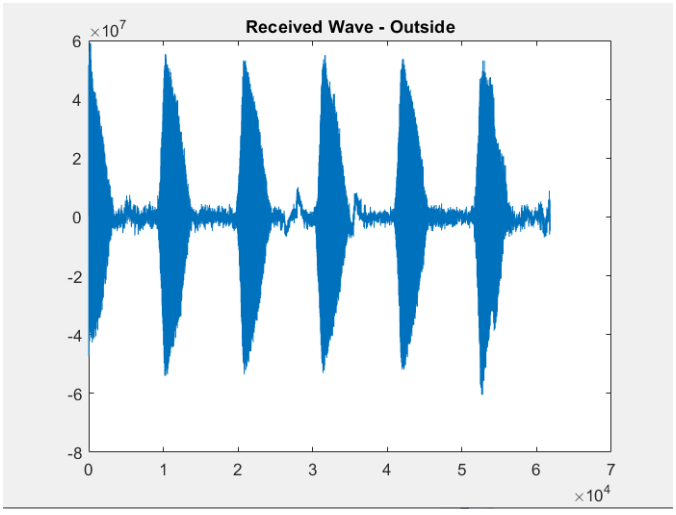
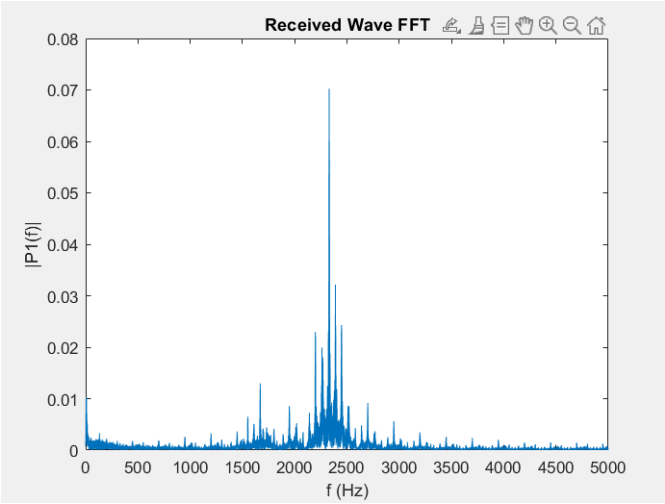


Figure 20: Electronic Architecture

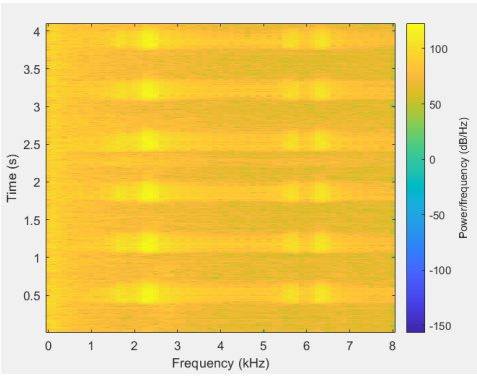
5.2 Testing



(a) Received Signal Outside



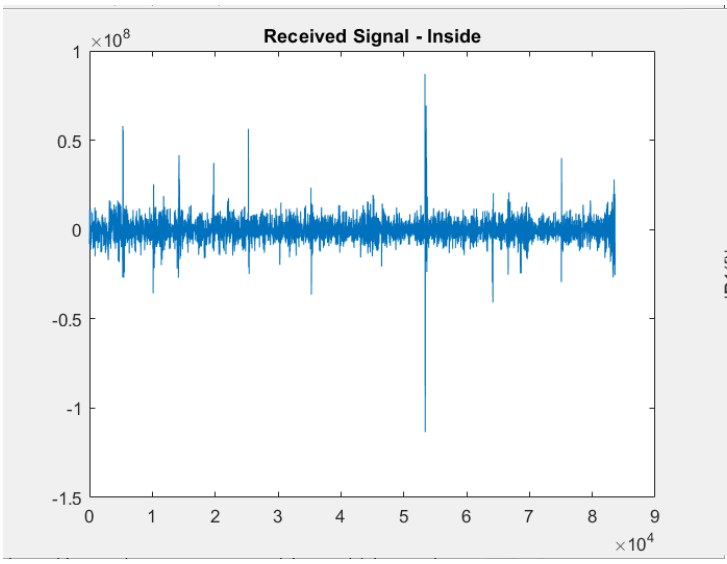
(b) Fourier Transform of Received Signal



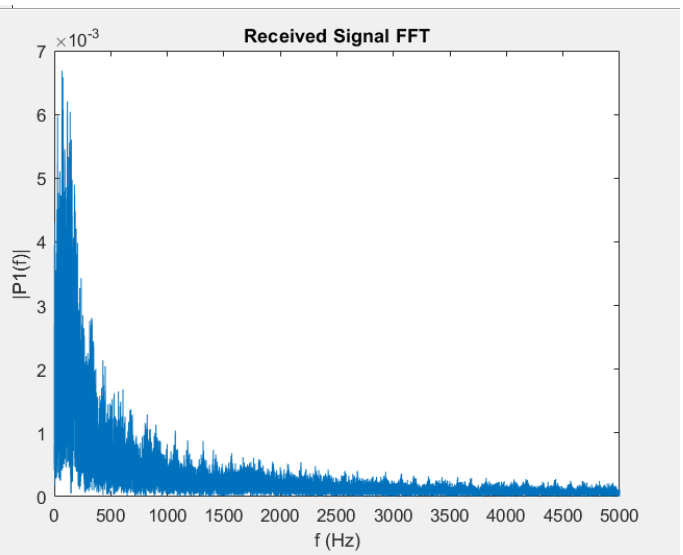
(c) Time Frequency Spectrogram

Figure 21: Received Signal Outside

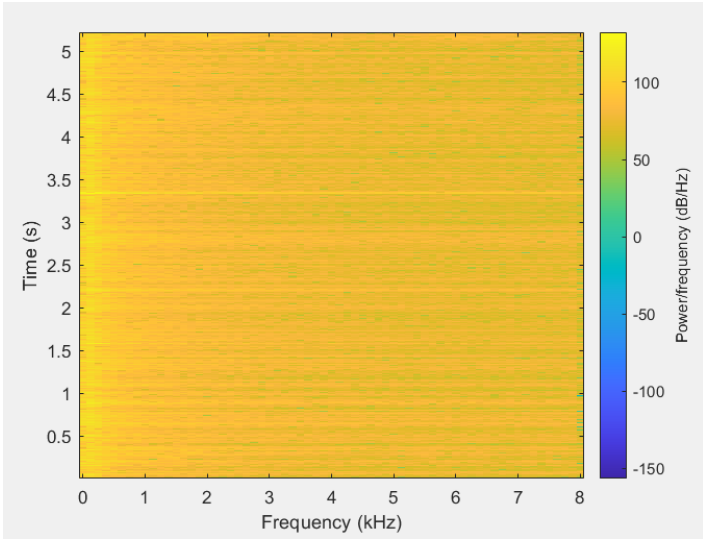
Testing has been conducted in the above pipe setup after assembling the required electronics. A Hanning windowed toneburst input signal of frequency 2500Hz and with 2500 cycles has been generated and played through the speaker. Initially the background noise has been recorded on the microphone. On receiving the signal, this background noise has been removed using MATLAB. Fourier analysis was performed on this signal for measuring the frequency distribution.



(a) Received Signal Inside



(b) Fourier Transform of Received Signal

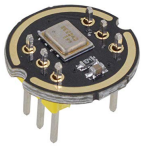


(c) Time Frequency Spectrogram

Figure 22: Received Signal Inside

Figure 21 shows the received signal with the speaker and the mike outside the pipe. Figure 22 shows the received signal with the speaker and mike submerged at both ends of the pipe. It can be observed that there is a lot of noise in the signal and even the Fourier analysis couldn't separate the frequency. Multiple rounds of testing were

performed with similar results. Another round of testing was performed by using a piezoelectric microphone instead of the INMP441 which is an MEMS microphone. This also failed to yield any positive results.



(a) INMP441 MEMS Microphone

(b) Piezoelectric Contact Microphone

Figure 23: Microphones Used

5.3 Discussion

Several practical issues were encountered while conducting the experiments. The major problem being that the microphones and speakers used were not specifically meant for underwater purposes. Special modifications were made to ensure they are kept underwater and this could have impacted their performance. Use of a hydrophone could yield better results. The pipe setup used also had lot of scaling and rust present. This could have also impacted the propagation of the waves inside the pipes. Further extensive testing with better suited equipment needs to be performed to effectively validate the results.

Chapter 6

Future Work

6.1 Testing

Further rounds of testing with suitable equipment has to be performed to validate the simulation results. Use of a hydrophone and an underwater capable pinger is a possible solution for better signal transmission. The testing can be conducted in the smaller pipe assembly until validation.

After validation in the smaller pipe assembly, the testing will move to a larger pipe assembly of length greater than 50m. This testing will focus on the reliability and accuracy of the solution in close to actual conditions. This might require use of industrial grade sensor equipment. The utility of the solution in case of impurities in the fluid, scaling in pipes, leaks and cracks in pipes etc will be tested in this phase. Based on the results from testing, suitable modifications will be made.

6.2 Additional Simulations

Along with testing, additional simulations have to be performed by incorporating real life inefficiencies and losses. These simulations will be validated in the final phase of testing.

6.3 Final Product Design

After both preliminary and final testing, the focus will be to develop the solution into a complete product. The predicted distance could be made more accurate by combining it with the dead reckoning data from IMUs. Based on the actual range of operation of acoustic waves, the final product will consist of multiple transmitters along the pipe length. As the robot passes through each of the transmitter, it receives the waves and calculates the distance from the reference location. The transmitters could be installed along with the pipe while the receiver is placed on the robot. This setup could track the robot in straight pipes for multiple kms.

Chapter 7

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